

Professional Installation Manual

Ubuntu Server 26.04 LTS Enterprise Deployment

Prepared for: ABC Startup Solutions

ITEP 414 – System Administration and Maintenance

Week 3 Portfolio Project | BSIT Program

Laguna State Polytechnic University

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1. Introduction

This manual documents the complete deployment of an Ubuntu Server 26.04 LTS virtual machine, written so that another system administrator could follow it without additional instructions. It covers installation, initial configuration, verification, troubleshooting, and best practices for maintaining the server going forward.

2. Ubuntu Server Installation Guide

2.1 Prerequisites

- Oracle VirtualBox (or VMware Workstation) installed on the host machine
- Ubuntu Server 26.04 LTS ISO downloaded from ubuntu.com/download/server
- Host machine with at least 8 GB RAM and 50 GB free disk space

2.2 Virtual Machine Specifications

Component	Value
Name	Ubuntu-Server-Week03
RAM	4096 MB (4 GB)
CPU	2 Virtual Processors
Storage	40 GB (VDI, dynamically allocated)
Network	NAT

2.3 Installation Steps

1. Create a new VM in VirtualBox and attach the Ubuntu Server ISO.
2. Select language: English. Select keyboard layout: English (US).
3. Accept DHCP for network configuration; record the assigned IP address.
4. Leave proxy configuration blank unless a proxy is required.
5. Accept the default Ubuntu archive mirror.
6. Choose "Use an entire disk" for guided storage configuration, with LVM enabled.
7. Set the hostname to server01.
8. Create a non-root administrative user with a strong password.
9. Enable "Install OpenSSH Server" with password authentication.
10. Skip optional server snaps unless specifically required.
11. Complete the installation and reboot when prompted.

3. Initial Server Configuration

After first boot, confirm the following configuration details match what was set during installation:

```
Hostname: server01
IP Address: 10.0.2.15/24 (DHCP via enp0s3)
SSH: OpenSSH Server enabled, password authentication allowed
```

Apply system updates immediately after first login:

```
sudo apt update
sudo apt upgrade -y
```

4. Server Verification Procedures

Step	Command	Expected Result
Login	—	Successful login with created user account
Hostname	hostname	server01
IP Address	ip addr	Matches DHCP-assigned address
Connectivity	ping -c 4 google.com	4 successful replies
Updates	sudo apt update && sudo apt upgrade -y	Packages fetched and installed
SSH Service	systemctl status ssh	active (running)

5. Troubleshooting Guide

5.1 Installer appears frozen during "installing kernel" step

Symptom: The installer log shows kernel watchdog "soft lockup" messages and appears to stop progressing.

Resolution: Open the full installer log and check whether new lines are still appearing over time. If the log is still updating, the process is working but slow — allow it to continue. Only power off and restart the VM if the log has been completely static for an extended period (20+ minutes) with no CPU activity.

5.2 "403 Forbidden" error during apt upgrade

Symptom: A specific package (e.g., an optional linux-firmware package) fails to download with a 403 Forbidden error from the regional mirror.

Resolution: Re-run `sudo apt update` followed by `sudo apt upgrade -y`. This is typically a transient mirror-side issue. If it persists on a non-critical package, it can be safely ignored, since the majority of updates will still install successfully.

5.3 SSH command not recognized

Symptom: `systemctl status ssh` returns "command not found."

Resolution: Check for typos — the correct command is `systemctl` (ending in "tl"), not `systemcti` or similar. Bash will suggest the correct command if a close match exists.

6. Best Practices

- Always verify each configuration step immediately after applying it, rather than assuming success.
- Use guided LVM partitioning on new servers to allow for easier disk expansion later.
- Enable SSH during installation rather than as a separate post-install step, to save time on remote-access setup.
- Avoid installing unnecessary packages or server snaps unless the server's specific role requires them.
- Document the exact hostname, IP address, and installed services immediately after setup for future reference.
- Re-run failed update commands before troubleshooting further — many package-mirror errors are transient.

7. References

Canonical Ltd. (2026). Ubuntu Server documentation. <https://ubuntu.com/server/docs>

Microsoft. (2026). Windows Server 2025 Evaluation Center. <https://www.microsoft.com/evalcenter>

Oracle Corporation. VirtualBox User Manual. <https://www.virtualbox.org/manual/>